

VINAY DEVABHAKTUNI

Charlotte, North Carolina | +1(509)338-8387 | vinaydevabhaktuni2028@gmail.com | [LinkedIn](#) | [GitHub](#)

PROFESSIONAL SUMMARY

AI Data Engineer with 2+ years building cloud-native ETL/ELT pipelines and analytics infrastructure. Production experience in Python, SQL, and PySpark designing partitioned data layouts, tuning complex queries, automating data quality validation, and integrating ML/LLM workflows with downstream BI and analytics consumers. Co-author, IEEE article on trust engineering for AI-generated code.

TECHNICAL SKILLS

- **Programming and Querying:** Python, Java Spring Boot, SQL, PL/SQL, GraphQL, JavaScript, Node, React, Angular
- **ETL and Orchestration:** Airflow, PySpark, Spark SQL, Spark Streaming, Spark ML, Event Driven ETL/ELT, REST/GraphQL Ingestion, Kafka
- **Data Modeling and Quality:** Partitioning, Indexing, Automated Data Validation, Query Optimization, Schema Enforcement
- **Cloud (AWS):** S3, Glue, Glue Data Catalog, Athena, Lambda, EC2, CloudWatch, IAM, SageMaker, Redshift
- **Big Data and Storage:** Hadoop, HDFS, Hive, MySQL, NoSQL stores
- **AI Integration:** LLM Evaluation, RAG, FAISS, SentenceTransformers
- **DevOps:** Git, GitHub, CI/CD, Docker, GitHub Actions, Jenkins
- **LLM's:** Claude, Gemini, ChatGPT, Ollama

PROFESSIONAL EXPERIENCE

University of North Carolina at Charlotte

Graduate Research Assistant

Charlotte, United States of America. · Aug 2024 - Present

- Architected a **serverless event-driven ETL pipeline** on AWS (S3 → Lambda → Glue Crawler → Athena → Flask/EC2), partitioning raw ingestion by date and source to reduce query scan and **power five self-service analytics queries for research stakeholders**.
- Built a **modular four-stage PySpark/Spark SQL transformation layer** with reusable feature modules and tested data contracts, processing multi-source structured and semi-structured inputs into curated analytical tables.
- **Automated multi-stage research workflows in Airflow + Lambda** under scoped IAM least-privilege roles with CloudWatch monitoring, **eliminating ~15 hrs/week of manual operations**.
- Deployed a **Flask service on EC2 to expose Athena query results** (over S3-partitioned data) as a stakeholder-facing analytics dashboard, completing the **serverless ingestion-to-presentation loop**.
- Co-author an IEEE article on trust engineering for LLM-generated code; define a five-class hallucination taxonomy (H1–H5) with programmatic grounding tests across GPT-4, Claude, and Codex, supporting ML model evaluation in production data pipelines

Cognizant Technology Solutions

Programmer Analyst

Chennai, India · Jun 2023 - Jun 2024

- **Migrated a high-traffic legacy monolith to Docker-containerized microservices on AWS**, reducing long-term technical debt and improving deployment cadence.
- **Tuned Spring Framework REST endpoints and backend SQL** (indexing, query plan analysis) to **cut API response times 30% under production load**.
- Established **GitHub Actions CI/CD quality gates running AI-assisted code review** and static analysis, accelerating integration and reducing post-deployment defects across the team.
- Improved distributed processing efficiency 20% by parallelizing Java algorithms with multithreading.

Programmer Analyst Trainee

Hyderabad, India · Aug 2022 - Jul 2023

- **Engineered Python ingestion pipelines with 10+ automated data validation workflows** (schema checks, null-rate thresholds, referential integrity), reducing manual reconciliation 40% while maintaining 99.9% data integrity.
- **Accelerated complex SQL query execution 25%** by refactoring with stored procedures, triggers, views, and partition-aware indexing across heterogeneous transactional sources.
- **Orchestrated end-to-end ETL workflows in Airflow, improving scheduling efficiency 20%**; deployed via GitHub CI/CD with rollback safeguards, cutting deployment errors 15%.
- Integrated Kafka real-time streaming into a Java Spring Boot microservices stack backed by MySQL and NoSQL stores, enabling sub-second processing latency for downstream services.
- Resolved critical production data and integration bugs in collaboration with senior engineers, improving pipeline reliability and observability.

PROJECTS

Manhua Matchmaker manhua-matchmaker.streamlit.app

Jan 2026 - Mar 2026

- Ingested ~10,000 records via MangaDex REST and AniList GraphQL APIs into a deduplicated catalog (600× speedup via fuzzy matching, 10 min → <1.5 s). Built semantic search over ~5,000 records with SentenceTransformers (MiniLM-L6-v2) + FAISS at sub-100 ms CPU latency, integrating ML embeddings into a user-facing query layer.

HUD FMR Fair Pay Dashboard github.com/VinayD2028/HUD-FMR-Dashboard

Feb 2026 - Apr 2026

- Co-built (3-person team) a Streamlit analytics application on HUD Fair Market Rents 2025, integrating county-level geographic, temporal, and housing-type data into stakeholder-facing visualizations - choropleth map, ranking views, pay fairness scan, and relocation calculator - translating public policy data into decision-ready analytics.

Job Search Automation github.com/VinayD2028/job-search-automation

Jan 2026 - Apr 2026

- Python 3.12 orchestration platform with Apify cron-based ingestion, SQLite six-state lifecycle tracking, Gmail API (OAuth2) inbox parsing, and local LLM (Ollama / llama3.2) for JD-aware tailoring end-to-end pipeline ownership from ingestion through ML inference.

EDUCATION

University of North Carolina at Charlotte <i>Master of Science, Computer Science</i> • GPA: 3.83/4.0	Aug 2024 - Present <i>Charlotte, NC</i>
CMR Technical Campus <i>Bachelor of Technology, Computer Science Engineering</i> • GPA: 7.24/10.0	Aug 2018 - Jul 2022 <i>Hyderabad, India</i>

CERTIFICATES

- AWS Academy Data Engineering: Completion Dec 2025
- Python Data Analytics (Coursera/issuer)
- Generative AI: Enhance your Data Analytics Career (Coursera/issuer)
- SQL Certification (Coursera/issuer)